

Calculate the following limit:

$$\lim_{x \rightarrow 1} \left(\frac{\sqrt{x} - 1}{x - 1} \right)$$

$$\Rightarrow \frac{\sqrt{1} - 1}{1 - 1} = \frac{0}{0} \quad \underline{\underline{\text{I.F}}}$$

L'Hôpital Rule:

$$\frac{d}{dx} (\sqrt{x} - 1) = \frac{1}{2\sqrt{x}}$$

$$\frac{d}{dx} (x - 1) = 1$$

$$\frac{1}{2\sqrt{x}} \Rightarrow \frac{1}{2\sqrt{1}} = \boxed{\frac{1}{2}}$$



Calculate the following limits:

$$\lim_{x \rightarrow 2} \frac{x^2 - 4}{\sqrt{x+2} - 2}$$

$$\Rightarrow \frac{2^2 - 4}{\sqrt{2+2} - 2} = \frac{4 - 4}{2 - 2} = \frac{0}{0} \quad \underline{\text{I.F.}}$$

L'Hôpital Rule:

$$\frac{d}{dx}(x^2 - 4) = 2x$$

$$\frac{d}{dx}(\sqrt{x+2} - 2) = \frac{1}{2\sqrt{x+2}}$$

$$\frac{\frac{2x}{1}}{2\sqrt{x+2}} = \frac{4x\sqrt{x+2}}{1} \Rightarrow \frac{4(2)\sqrt{2+2}}{2\sqrt{4}} = \frac{8(2)}{2\sqrt{4}} = \boxed{16}$$

$$\lim_{x \rightarrow 2} \frac{x^2 - 4}{\sqrt{x+2} - 2} = 16$$



Calculate the following limit:

$$\lim_{x \rightarrow 1} \frac{x^3 - 1}{x^2 - 1}$$

$$\Rightarrow \frac{1^3 - 1}{1^2 - 1} = \frac{0}{0} \text{ I.F.}$$

L'Hôpital Rule:

$$\frac{d}{dx}(x^3 - 1) = 3x^2$$

$$\frac{d}{dx}(x^2 - 1) = 2x$$

$$\Rightarrow \frac{3x^2}{2x} = \frac{3(1)^2}{2(1)} = \boxed{\frac{3}{2}}$$

$$\lim_{x \rightarrow 1} \frac{x^3 - 1}{x^2 - 1} = \frac{3}{2}$$



calculate the following limit:

$$\lim_{x \rightarrow \infty} \left(\sqrt{x^2 + 3x + 1} + x \right)$$

$$\frac{(\sqrt{x^2 + 3x + 1} + x)(\sqrt{x^2 + 3x + 1} - x)}{(\sqrt{x^2 + 3x + 1} - x)}$$

$$\frac{x^2 + 3x + 1 - x^2}{\sqrt{x^2 + 3x + 1} - x} = \frac{3x + 1}{\sqrt{x^2 + 3x + 1} - x} = \frac{3x + 1}{\sqrt{x^2 \left(1 + \frac{3}{x} + \frac{1}{x^2} \right)} - x}$$

$$\Rightarrow \frac{3}{x} = 0 \text{ and } \frac{1}{x^2} = 0$$

$$\text{So: } \frac{3x + 1}{\sqrt{x^2(1 + 0 + 0)} - x} = \frac{3x + 1}{\sqrt{x^2} - x}$$

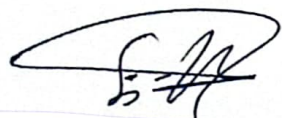
$$\Rightarrow \sqrt{x^2} = -x \Rightarrow x \rightarrow -\infty$$

then:

$$\frac{3x + 1}{-2x} = \frac{-\infty}{\infty} \text{ I.F.}$$

L'Hôpital Rule:

$$\boxed{\frac{3}{-2}} \Rightarrow \lim_{x \rightarrow -\infty} (\sqrt{x^2 + 3x + 1} + x) = \frac{3}{-2}$$



$$\lim_{x \rightarrow \infty} \left(1 + \frac{3}{x}\right)^{2x}$$

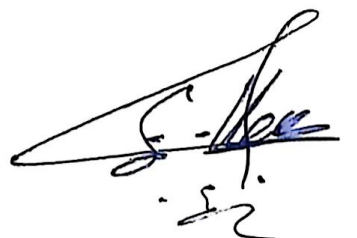
$$\ln \left(1 + \frac{3}{x}\right)^{2x} = 2x \cdot \ln \left(1 + \frac{3}{x}\right)$$

$$\ln \left(1 + \frac{3}{x}\right) \approx \frac{3}{x}$$

$$\Rightarrow 2x \cdot \frac{3}{x} = \frac{6x}{x} = 6$$

then: e^6

$$\lim_{x \rightarrow \infty} \left(1 + \frac{3}{x}\right)^{2x} = e^6$$



$$\lim_{x \rightarrow \infty} \left(1 + \frac{5}{x}\right)^{3x}$$

$$\ln \left(1 + \frac{5}{x}\right)^{3x} = 3x \ln \left(1 + \frac{5}{x}\right)$$

$$\Rightarrow \ln \left(1 + \frac{5}{x}\right) \sim \frac{5}{x}$$

$$\text{then: } 3x \cdot \frac{5}{x} = \frac{15x}{x} = 15$$

$$\Rightarrow e^{15}$$

So:

$$\lim_{x \rightarrow \infty} \left(1 + \frac{5}{x}\right)^{3x} = e^{15}$$



$$\lim_{x \rightarrow \infty} \left(\frac{1 + \frac{4}{x}}{1} \right)^{2x+3}$$

$$\ln \left(1 + \frac{4}{x} \right)^{2x+3} = (2x+3) \cdot \ln \left(1 + \frac{4}{x} \right)$$

$$\ln \left(1 + \frac{4}{x} \right) \approx \frac{4}{x}$$

$$\Rightarrow (2x+3) \cdot \frac{4}{x}$$

$$\Rightarrow \frac{8x+12}{x}$$

then: ~~$\frac{\infty}{\infty}$~~ $\frac{\infty}{\infty}$ I.F

L' Hospital Rule

$$\Rightarrow 8 \quad \text{then: } e^8$$



$$\lim_{x \rightarrow \frac{\pi}{4}} \frac{\tan^3 x - 1}{2 \sin^2 x - 1}$$

$$\tan\left(\frac{\pi}{4}\right) = 1$$

$$\sin\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$$

$$\Rightarrow \frac{1^3 - 1}{2\left(\frac{1}{\sqrt{2}}\right)^2 - 1} = \frac{1 - 1}{\frac{2}{2} - 1}$$

then: $\frac{0}{0}$ I.F

L'Hôpital Rule:

$$\frac{d}{dx}(\tan^3 x - 1) = 3 \tan^2 x \cdot \sec^2 x$$

$$\frac{d}{dx}(2 \sin^2 x - 1) = 4 \sin x \cdot \cos x$$

$$\Rightarrow \frac{3 \tan^2 x \cdot \sec^2 x}{4 \sin x \cdot \cos x} = \frac{3(1^2 \cdot 2)}{4\left(\frac{1}{\sqrt{2}}\right) \cdot \frac{\sqrt{2}}{2}}$$

$$\frac{3 \cdot 2}{\frac{4}{\sqrt{2}} \cdot \frac{\sqrt{2}}{2}} = \frac{6}{\frac{4\sqrt{2}}{2\sqrt{2}}} = \frac{6}{2} = 3$$

$$\text{So: } \lim_{x \rightarrow \frac{\pi}{4}} \frac{\tan^3 x - 1}{2 \sin^2 x - 1} = 3$$



$$\lim_{x \rightarrow \infty} \left(1 + \frac{6}{x} \right)^{2x-5}$$

$$\ln \left(1 + \frac{6}{x} \right)^{2x-5} = (2x-5) \ln \left(1 + \frac{6}{x} \right)$$

$$\Rightarrow \ln \left(1 + \frac{6}{x} \right) \approx \frac{6}{x}$$

$$\text{Then: } 2x-5 \left(\frac{6}{x} \right) = \frac{12x-30}{x} = \frac{\infty}{\infty}$$

L'Hôpital Rules:

$$\frac{d}{dx}(12x-30) = 12$$

$$\Rightarrow 12$$

$$\frac{d}{dx}(x) = 1$$

$$\text{So: } e^{12}$$

$$\lim_{x \rightarrow \infty} \left(1 + \frac{6}{x} \right) = e^{12}$$

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